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# Application of Q-Band Electron Spin Echo Spectrometer to Investigation of Glasses Doped with Rare Earth Ions

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**Abstract.** The Q-band electron spin echo (ESE) spectrometer which was created using modern microwave components is described. This simple incoherent apparatus was used with the X-band one for the study of phosphate and silicate glasses doped with non-Kramers rare earth $Tb^{3+}$ ions. The EPR spectra measured by the ESE method have frequency independent peaks. The experimental results presumably show the existence of several types of paramagnetic centers in studied systems.

## 1. Introduction

The electron spin echo (ESE) method is known for a rather long time, but only in the last ten years it has been undergoing rapid development [1]. This advance has been caused by the progress of digital electronics and microwave techniques and by the vast possibilities of the ESE method. Some advantages of the ESE method [2] may be pointed out.

ESE method allows one to detect absorption EPR spectra instead of the first derivative ones as in ordinary continuous wave (CW) EPR. This gives the possibillity to avoid a loss of information while registering very broad inhomogeneous lines, for example in glasses doped with rare earth (RE) ions. The valuable feature of the ESE is the possibility to distinguish between different types of paramagnetic centers according to their relaxation times.

This method allows one to carry out the investigation of dynamic properties of paramagnetic centers by measuring spin-lattice and phase relaxation times.

Finally, one of the most important applications of the ESE method is investigating the structure of various solids by analyzing the modulation of echo decay envelopes.

The overwhelming majority of ESE spectrometers operate at the X-band. Meanwhile, as early as 1965 Mims noticed in his paper [3] the great advantages of the operation at higher frequencies. The nearest to the X-band a high frequency range (where microwave components are also well developed) is the Q-band. Besides well known advantages of the Q-band as compared with the X-band in CW EPR such as higher sensitivity and resolution, the ESE method has its own advantages.

One of them is the possibility of using lower power and hence cheaper solid state microwave generators as compared with TWT and magnetrons. This is because the volume of a standard resonator at the Q-band is considerably less than that at the X-band and therefore it is easier to reach the necessary amplitude of microwave (MW) magnetic field $B_1$, the same pulse duration and MW power provided.

Another valuable feature of the Q-band ESE is the possibility to investigate weak hyperfine interactions which are difficult to be measured at the X-band. For instance, centers possessing high effective $g$-value may have the inconveniently deep modulation of the envelope at the low fields needed for the X-band operation [2].

Finally, the Q-band ESE may be used as a supplement instrument to the X-band one for investigations of various frequency dependences.

All these reasons impelled us to construct simple incoherent Q-band spectrometer using modern achievements at microwaves.

## 2. Experimental Set-Up

The block diagram of the Q-band bridge is presented in Fig. 1.

Transmitter and receiver units are the main parts of the bridge. The transmitter unit consists of three generators.

1. High power pulse generator on IMPATT diodes with 2 W output power is designed for 2-pulse and 3-pulse echo sequences. The power is controlled simultaneously for all MW pulses using the electronic attenuator of 30 dB.
2. Low power high stability generator on Gunn diode with 50 mW output power serves for tuning and observation of EPR. Power of MW generator is adjusted by electrically controlled attenuator of 30 dB.
3. Low power generator on Gunn diode for long pulses (output power is 50 mW). It serves for pulse saturation method.

MW frequency of 35.4 GHz is fixed in all regimes. The MW pulses from the generator unit are fed through the circulator and the matching transformer

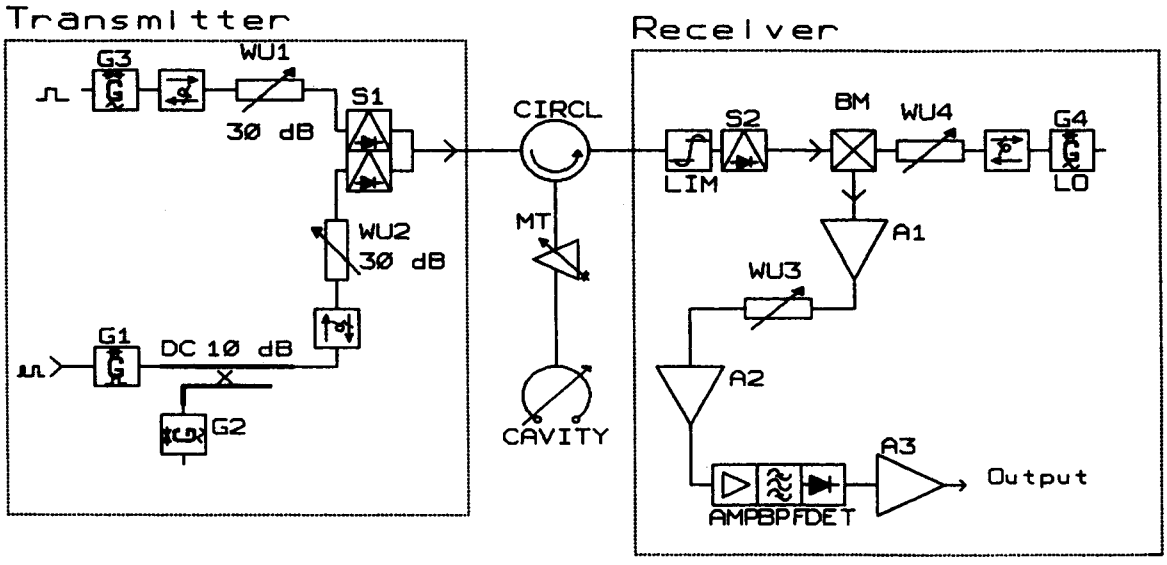



**Fig. 1.** The block diagram of the Q-band electron spin echo bridge: G1 – the high power pulse generator (two IMPATT diodes), G2 – the high stability CW generator (Gunn diode), G3 – the long pulse generator (Gunn diode), G4 – the heterodyne, WU1–WU3, WU5 – the electrically controlled attenuators 30 dB, WU4 – the tunable attenuator, S1 – the channel switch (SWTC), S2 – the protection switch (SWT), MT – the matching transformer, LIM – the limiter, BM – the balanced mixer, A1, A2 – the intermediate frequency amplifiers 30 dB, A3 – the video amplifier, AMP – the matching amplifier, BPF – the band pass filter, DET – the video detector,

to a resonator. The signal from the resonator through the same circulator is directed to the receiver unit. This unit represents the low-noise superheterodyne receiver with one fold down frequency conversion. The intermediate frequency (IF) is 1.6 GHz, which is commonly used in up-to-date microwave receivers. This allows us to operate with short pulse signals ($\tau \geqq 10$ ns), and to reduce requirements to the heterodyne noise. The waveguide-integrated balanced mixer (BM) with the conversion loss of 5.6 dB serves as the main part of the receiver. It is made by means of the "finline" technology. The maximum possible MW power that can be fed to the mixer diodes, is rather low, hence to avoid their destroying the rapid protector limiter (LIM) is set at the input. This LIM limits the MW power on the level of 10 mW. In addition the fast switch S2 (isolation of 40 dB) protects the receiver input during the high power pulses. The control signal is fed to the switch from the device that initiates these pulses. The limiter and the switch keep the IF receiver (A1, A2) near the saturation point and reduce its paralysis time to a minimum. A1, A2 are microstrip three stage bipolar transistor 30 dB amplifiers with the noise figure of 2.6 dB. Since the dynamic range of the receiver is limited by the maximum output power, the electrically controlled attenuator WU3 of 30 dB is used for the operation at high level signals (for example in CW regime). The output signal from A2 is fed to the microstrip unit M which consists of a matching amplifier (AMP), a band pass filter (BPF) and a video detector (DET). The filter pass band is chosen by taking into account a really accessible minimum of a pulse duration of the micro-

wave generator. The time constant of the detector is rather small. The video amplifier has a small gain, a broad pass band and the output resistance of 50 Ω.

The option of regime, variation of MW power in the generator unit and amplification in the receiver unit can be done by MW controller.

The magnetic field is supplied by the usual electromagnet up to 1 T and superconducting coils up to 2.5 T. The pumping system allows the operation at liquid helium temperatures down to 1.5 K.

The X-band ESE spectrometer was described earlier [4] and represents incoherent set-up with a magnetron transmitter and the TWT receiver.

## 3. Object of Investigation

This Q-band ESE spectrometer was used with the X-band one for the study of phosphate and silicate glasses doped with non-Kramers rare earth $Tb^{3+}$ ions.

Although we present here some preliminary results of the study they demonstrate convincingly the usefulness of the Q-band ESE spectrometer.

As it is well known EPR of non-Kramers ions is relatively rarely observed in crystals. Theoretical calculations show that non-Kramers ions may have degenerate states only in sites with relatively high local symmetry of the crystal field. In fact, EPR within such non-Kramers doublet was reported to be found in crystals with threefold and fourfold axial symmetry sites [5]. In sites with low symmetry of the crystal field the ground level of the free ion is split into singlets. The energy separation between these singlets may be too large as compared with an ordinary EPR microwave quantum and in these cases EPR is not observed. However, there are some works in which EPR in the low symmetry centers was found [6–8]. In glasses the impurity centers were expected to have sites of a low symmetry, but the situation here must be essentially more complex than in crystals. So reports about the studies of EPR in glasses doped with non-Kramers rare earth ions are of great interest. At present only a few papers about such observations were published. It should be underlined here that except a few cases such as S-state $Gd^{3+}$ ions, the study of glasses doped with RE ions by ordinary EPR methods is a very difficult task in general. The reason is that RE ions in such hosts have very broad inhomogeneous EPR lines where ordinary CW EPR is little informative [9]. Besides short spin-lattice relaxation times lead to the necessity to carry out measurements at liquid helium temperatures. The first observation of EPR of some non-Kramers ions in phosphate glasses was reported by A.A. Antipin and V.S. Zapasskii in 1982 [10]. They used the laser Faraday rotation method. However they did not give any explanation or models.

Later there appeared some doubts about the nature of the detected signals [11]. In 1988 and 1991 Harris and Furniss reported about the observation of the relatively well resolved EPR spectra in the fluorozirconate glass doped with $Tb^{3+}$ and $Tm^{3+}$ ions [12, 13]. They also gave a model of the rare earth ion site in that glass. Taking into account the abovementioned advantages of the ESE method we have investigated EPR of $Tb^{3+}$ ions in two quite different oxide glasses supposing that these ions may be good probes for a symmetry site definition in glasses.

The investigated samples are optical glasses, synthesized at Vavilov State Optical Institute in St. Petersburg. They were prepared from especially pure reagents. These are oxide glasses with different network formers and modifiers. The chemical formulae of glasses are:

$75P_2O_5 \cdot 25La_2O_3$ (phosphate glass),

$59.5SiO_2 \cdot 20Na_2O \cdot 20B_2O_3 \cdot 0.5La_2O_3$ (silicate glass).

RE impurity was introduced as oxide of terbium ions $Tb_2O_3$ instead of $La_2O_3$ with a concentration of 0.1 mol %.

## 4. Experimental Procedure and Results

Measurements of the EPR spectra were made using ESE set-ups with working frequencies of 9.35 and 35.4 GHz in the temperature range 1.6—4.2 K. Experiments were carried out at the configuration of the magnetic field $B_0 \parallel B_1$ that differs from $B_0 \perp B_1$ one used in all previous measurements in glasses.

A reconstruction of the EPR spectra was made by the analysis of two pulse echo decays. Fig. 2 shows the typical echo decay for the silicate glass at the Q-band, that does not differ considerably in a shape from that at the X-band (for the phosphate glass echo decays are rather similar in the appearance to the silicate one). Although there is some deviation of decays from a pure exponent, which must be a straight line on this diagram, it has not essentially influenced a reconstructed shape of the EPR line because such a decay remains the same in the whole range of the magnetic field.

Measured experimental absorption EPR spectra are presented in Figs. 3—6.

The EPR spectrum of the phosphate glass doped with $Tb^{3+}$ at the X-band is shown in Fig. 3. Two well resolved peaks can be seen in it. The first maximum lies in the zero field, the second maximum is approximately at 370 G. The resonance line for the same glass at the Q-band is drawn in Fig. 4. There are also two well resolved peaks here but no strong absorption in zero fields is observed.

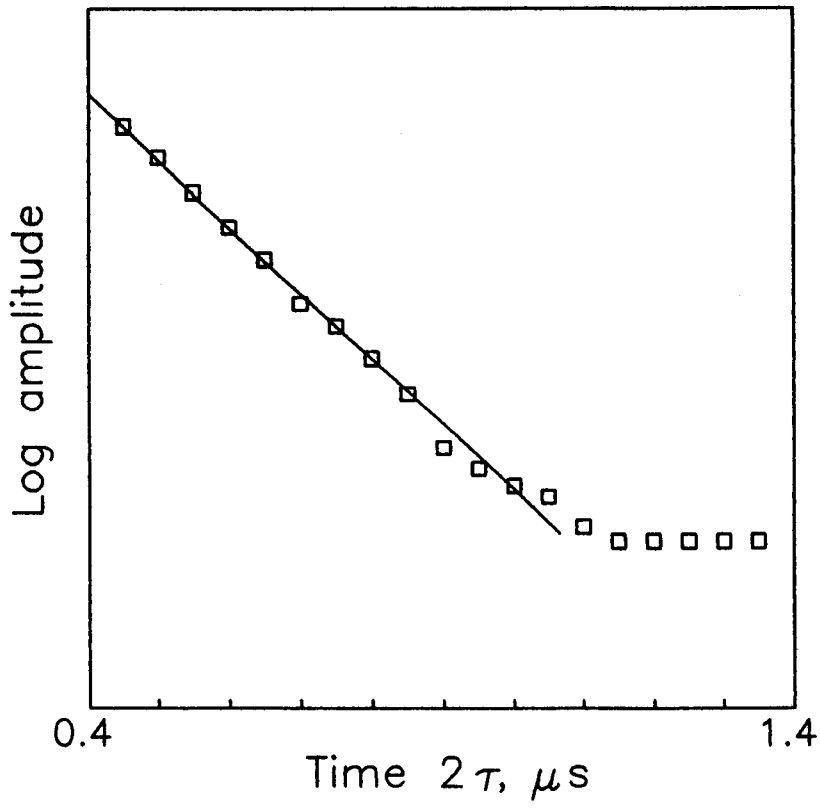



**Fig. 2.** The 2-pulse echo decay for $Tb^{3+}$ in the silicate glass at the Q-band (35.4 GHz), $T = 1.7$ K, $\tau$ is the time interval between the microwave pulses. The straight line corresponds to a pure exponent.

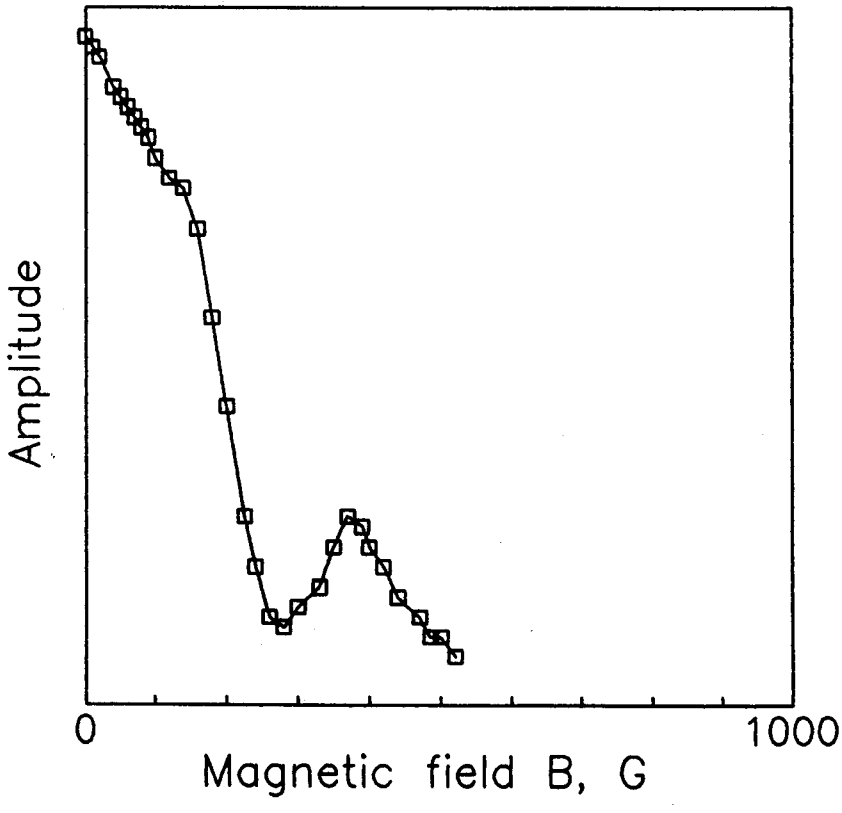



**Fig. 3.** The EPR spectrum of $Tb^{3+}$ in the phosphate glass by the ESE at the X-band (9.35 GHz), $T = 1.65$ K, $B_0 \parallel B_1$. Solid line connects experimental points.

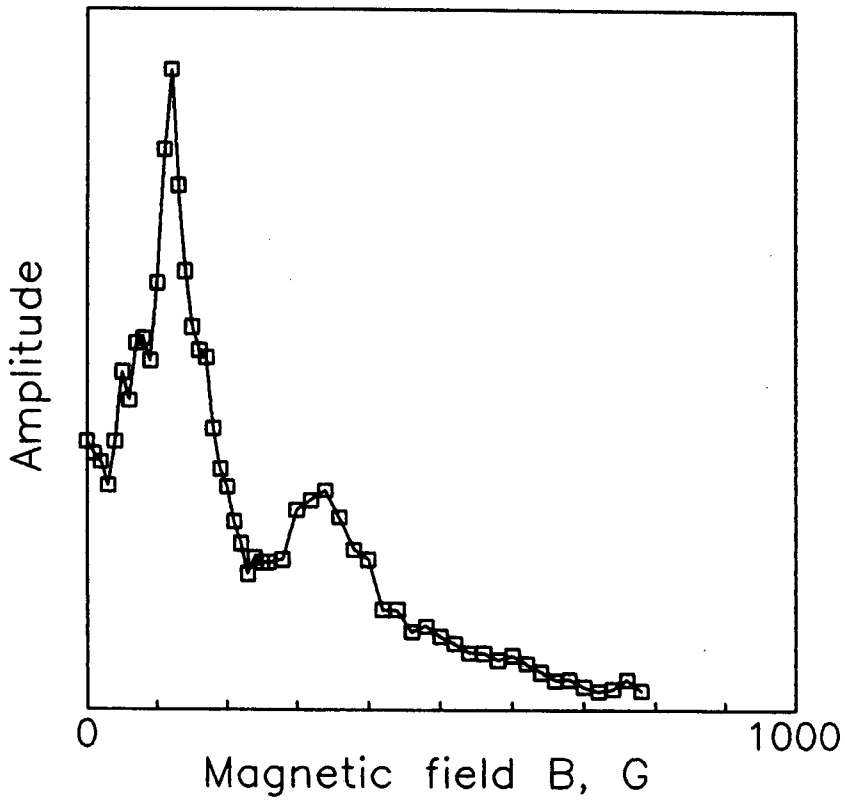



**Fig. 4.** The EPR spectrum of $Tb^{3+}$ in the phosphate glass by the ESE at the Q-band (35.4 GHz), $T = 2.0$ K, $B_0 \parallel B_1$. The solid line connects experimental points.

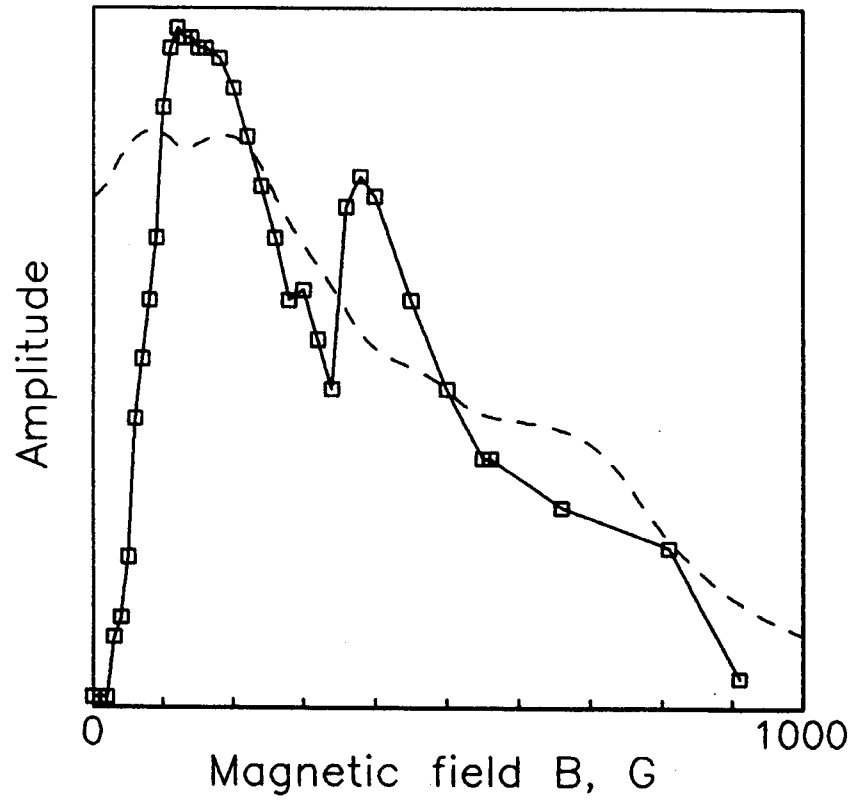



**Fig. 5.** The EPR spectrum of $Tb^{3+}$ in the silicate glass by the ESE at the X-band (9.35 GHz), $T = 1.65$ K, $B_0 \parallel B_1$. The solid line connects experimental points. The dashed line — the calculated EPR spectrum according to the model by Harris and Furniss [13] at 9.35 GHz, $g_\parallel = 17.3$, $A_\parallel = 0.202$ cm$^{-1}$. The distribution function $P(\delta) \sim \delta$. The configuration $B_0 \parallel B_1$.

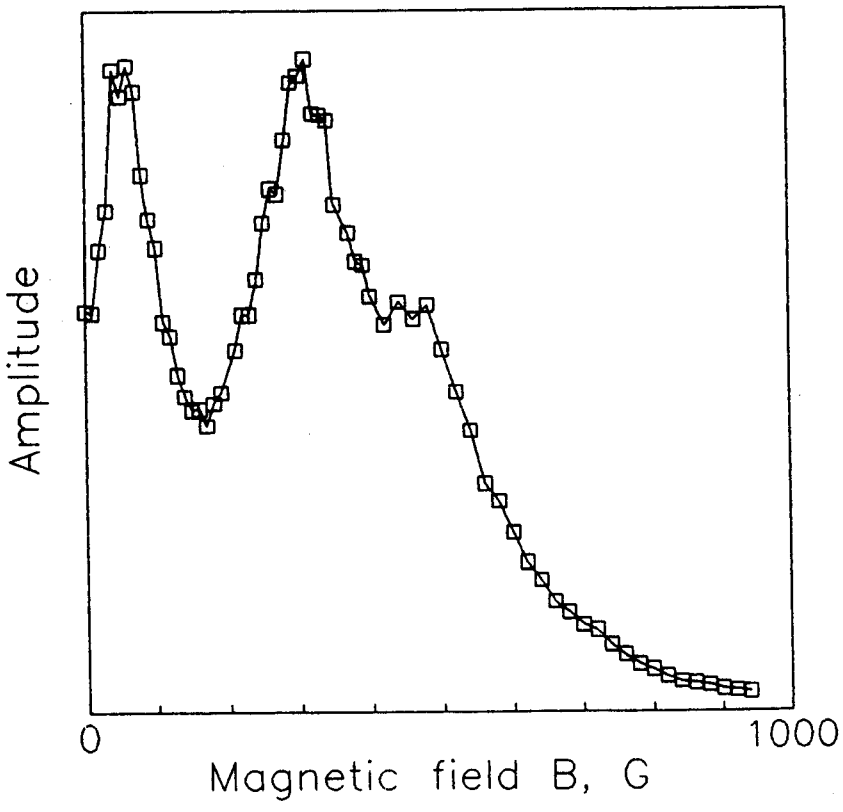



**Fig. 6.** The EPR spectrum of $Tb^{3+}$ in the silicate glass by the ESE at the Q-band (35.4 GHz), $T = 1.7$ K, $B_0 \parallel B_1$. The solid line connects experimental points.

Figure 5 shows the EPR spectrum for the silicate glass doped with $Tb^{3+}$ at the X-band (solid line). The picture here is similar to that in the phosphate glass, that is there are two peaks approximately at the same magnetic field values. It should be noted that there are sensible ESE signals at zero fields, though it is not seen clearly in the Figure. Fig. 6 presents the experimental results for the same glass at the Q-band. The EPR spectrum is essentially different here from that shown above. Two main peaks are observed at other field positions which are shifted to lower ones. There is also "a shoulder" in the field of 400–500 G.

## 5. Discussion

We have tried to interpret our experimental results using the model by Harris and Furniss [12, 13], that was developed for non-Kramers rare earth ion sites in the fluorozirconate glass. According to this model non-Kramers ions of $Tb^{3+}$ occupy a relatively well defined single type of site with a dominant crystal field term of axial symmetry, but with a continuous range of small site-to-site variations. The ground state is a non-Kramers doublet described by a well known spin Hamiltonian [5]

$$\mathcal{H} = g_{\parallel}\mu_B B_z S_z + \delta S_x + A_{\parallel} I_z S_z, \quad (1)$$

where the axis of quantization $z$ for each ion is in random directions. Parameters of $g_\parallel$ and $A_\parallel$ are taken as constants from site to site and have approximately the same values as they have in crystals [5]: $g_\parallel = 17.5 \pm 0.2$ corresponding to $|M_J\rangle = |\pm 6\rangle$; $A_\parallel = 0.203 \pm 0.006$ cm$^{-1}$. The zero-field splitting parameter $\delta$ takes a continuous range of values from 0 to at least 1 cm$^{-1}$. Harris and Furniss made calculations for two types of a distribution function for the parameter $\delta$: $P(\delta) = 1$ and $P(\delta) \sim \delta$ and determined that the latter fits their results better.

The energy levels corresponding to the spin Hamiltonian Eq.(1) are described by the equation

$$E_{1,2} = \pm \frac{\sqrt{\delta^2 + (g_\parallel \mu_B B_z + A_\parallel m)^2}}{2}, \qquad (2)$$

where $B_z = B_0 \cos\vartheta$, $\vartheta$ is the angle between the direction of the magnetic field $B_0$ and the $z$-axis; $m = \pm 3/2, \pm 1/2$. The energy level dependence Eq.(2) on $B_0$ has been plotted in Fig. 7. The dashed lines show the transitions which occur at $h\nu = \delta$ ($h\nu$ is a microwave quantum) and correspond to the characteristic values of the magnetic field: 1 denotes $B_z = A_\parallel / 2 g_\parallel \mu_B$, 2 denotes $B_z = 3A_\parallel / 2 g_\parallel \mu_B$.

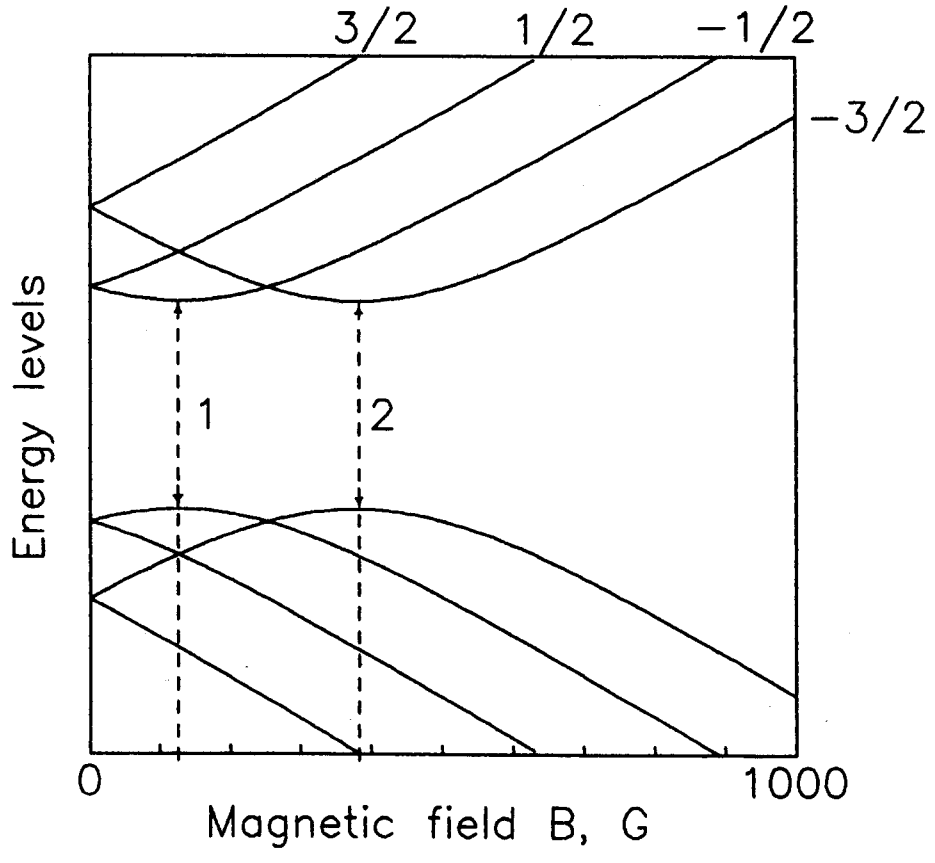



**Fig. 7.** The $Tb^{3+}$ ion energy level behavior vs. magnetic field, plotted according to the Eq.(2), with parameters $g_\parallel = 17.5$, $A_\parallel = 0.203$ cm$^{-1}$, $\delta = 0.1$ cm$^{-1}$. 1: $B_z = A_\parallel / 2 g_\parallel \mu_B$, 2: $B_z = 3A_\parallel / 2 g_\parallel \mu_B$.

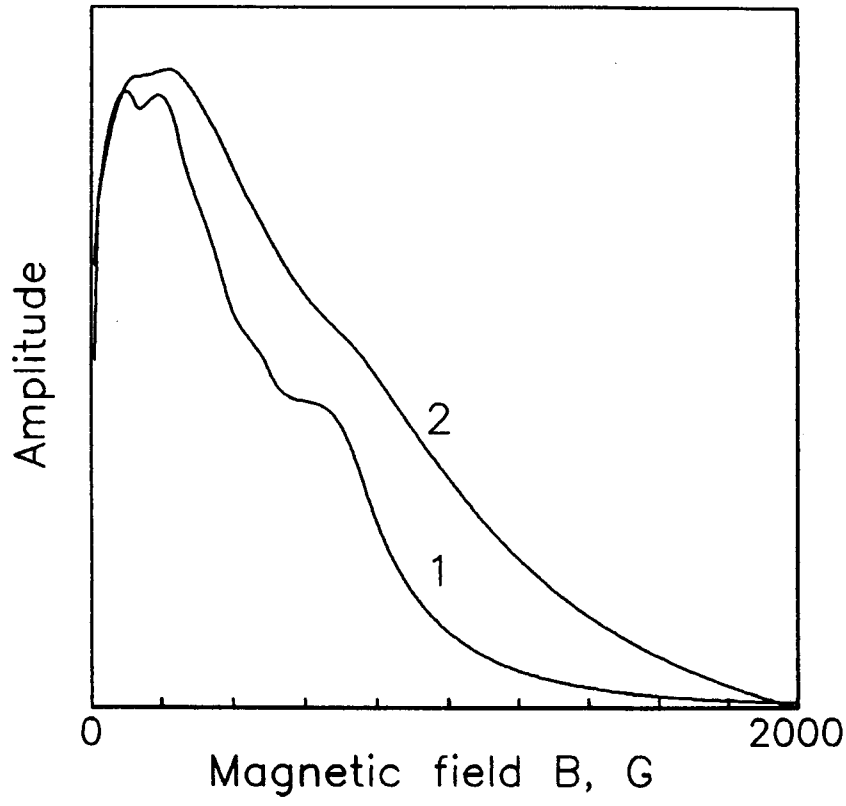



**Fig. 8.** EPR spectra of $Tb^{3+}$ ions calculated according to the model by Harris and Furniss at 9.35 GHz, $g_{\parallel} = 17.8$, $A_{\parallel} = 0.203$ cm$^{-1}$. The distribution function $P(\delta) \sim \delta$. 1 – $B_0 \parallel B_1$, 2 – $B_0 \perp B_1$.

We made calculations in accordance with those of Harris and Furniss taking necessary corrections for our condition $B_0 \parallel B_1$ instead of $B_0 \perp B_1$. It should be noted that the configuration $B_0 \parallel B_1$ is better than $B_0 \perp B_1$ because in the first case the EPR linewidth must be essentially narrower as compared with the latter and this was tested by our measurements. Fig. 8 shows the EPR spectra calculated according to Harris and Furniss for $B_0 \parallel B_1$ and $B_0 \perp B_1$ configurations. The broadening of the EPR spectrum and the disappearance of the features at the perpendicular configuration can be seen.

In Fig. 5, the EPR spectrum calculated according to Harris and Furniss for $B_0 \parallel B_1$ at the X-band (dashed line) is compared with our experimental EPR spectrum of $Tb^{3+}$ in the silicate glass (solid line). The fitting of $g_{\parallel}$ and $A_{\parallel}$ parameters has not essentially changed EPR spectrum shape. The discrepancy is more substantial for the phosphate glass at the X-band. At last our Q-band spectra completely disagree with the spectra calculated by Harris and Furniss. All this testifies that the discussed model cannot be used for treating our experiments in phosphate and silicate glasses doped with $Tb^{3+}$ ions.

Positions of the X and Q-band experimental peaks of the EPR spectra in the phosphate glass and of the X-band ones in the silicate glass correspond to the characteristic values of the magnetic field shown in Fig. 7, which may be observable at $h\nu = \delta$. In the silicate glass at the Q-band we observe quite

a different picture. The peak positions of the EPR spectrum were shifted to lower fields corresponding to the centers with $\delta$ value somewhat smaller than the microwave quantum value.

We suppose that the observed peculiarities in the behavior of the EPR spectra for $Tb^{3+}$ in the phosphate glass at two different frequencies can be explained by the existence of at least two types of paramagnetic centers. For the silicate glass there are at least three types of paramagnetic centers. Each of them should have its own nonmonotonic distribution of the zero-field splitting parameter $\delta$. The strong cross relaxation we found in the spin-lattice relaxation of $Tb^{3+}$ ions in the silicate glass is the indirect evidence of the existence of different type centers in the investigated glasses. However, the further experiments and development of the theory are needed for final conclusions to be drawn.

## 6. Conclusion

We presented here the preliminary results of the study of phosphate and silicate glasses doped with non-Kramers terbium ions by the Q- and X-band ESE. We suppose that we found at least two different types of paramagnetic centers in the phosphate glass and at least three types of sites in the silicate glass, each of them having its own nonmonotonic distribution of the zero-field splitting parameter $\delta$.

The obtained experimental results have convincingly demonstrated the usefulness of the Q-band ESE spectrometer.

## Acknowledgements

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